MUB: An Anonymized Benchmark for Evaluating Multimodal Large Language Models on Music Structure Understanding

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Abstract

We present the **Music Understanding Benchmark (MUB)**, a benchmark for evaluating the ability of multimodal large language models (MLLMs) to perceive and understand music at the level of a trained music analyst. MUB comprises 500 music pieces annotated by professional music annotators with ground-truth labels for four tasks: tempo estimation (BPM), key and tonality identification, time signature detection, and structural segmentation. A critical design feature of MUB is *full anonymization*: no original titles, artist names, or platform identifiers are included, preventing models from retrieving metadata via title-based lookup and ensuring that evaluation measures genuine audio understanding rather than memorized title-to-metadata associations. We provide two reference baselines: an objective DSP baseline that extracts features directly from waveforms (spectral flux for BPM, chroma-based Krumhansl–Schmuckler for key, beat periodicity for time signature), and a majority-vote baseline that simulates a text-only model with no audio access. End-to-end evaluation confirms the pipeline is functional: the majority baseline achieves 9.0% BPM accuracy, 9.2% key accuracy, 98.4% time signature accuracy (due to 4/4 dominance), and 0.000 structure F1, establishing the trivial lower bound that any audio-understanding model must exceed. We release the benchmark data, evaluation code, and baselines as an open repository.

1 Introduction

The rapid development of multimodal large language models (MLLMs) has expanded their input modalities beyond text and images to include audio. Models such as Qwen-Audio [2], SALMONN [3], and Gemini [4] can now accept audio inputs and generate text responses about what they "hear." However, evaluating whether these models truly *understand* music—as opposed to performing shallow pattern matching or leveraging memorized metadata—remains an open challenge.

Music understanding is qualitatively different from general audio understanding. It requires perceiving hierarchical temporal structures (beats, bars, phrases, sections), recognizing tonal frameworks (key, mode, harmonic function), and identifying formal segmentation (verse, chorus, bridge, etc.). These are the competencies that a trained music analyst applies when annotating a piece, and they are precisely the competencies that current benchmarks fail to test adequately.

Existing music understanding benchmarks have notable limitations. MuChoMusic [1] evaluates audio-language models with multiple-choice questions but relies on publicly available datasets where music titles are known, creating a potential leakage path: a model that has memorized metadata associations (e.g., "Bohemian Rhapsody is in B♭ major") can answer correctly without listening to the audio. Furthermore, MuChoMusic focuses on broad musical knowledge (genre, mood, instruments) rather than the structural and analytic tasks that professional annotators perform.

We address these gaps with the **Music Understanding Benchmark (MUB)**, which makes the following contributions:

1. **Four structural understanding tasks** grounded in professional annotation practice: BPM estimation, key/tonality identification, time signature detection, and structural segmentation (Section 4).
2. **Full anonymization**: no original titles, artist names, or platform identifiers (including internal music IDs and task IDs) appear in the benchmark. Each piece is assigned an opaque identifier (MUB-00001 to MUB-00500), eliminating the title-retrieval shortcut (Section 3.4).
3. **Expert-verified gold labels**: all ground truth is derived from professional music annotators' final submissions, cross-validated across multiple annotators and multiple listening passes.
4. **Two reference baselines**: an objective DSP baseline (waveform-level feature extraction) and a majority-vote baseline (text-only lower bound), enabling researchers to distinguish perception ability from prior knowledge (Section 6).
5. **Open evaluation pipeline**: Python code for prompt generation, prediction scoring, and baseline reproduction, released as an independent repository.

2 Related Work

2.1 Music Information Retrieval

Classical MIR tasks—BPM estimation [5], key detection [6], chord recognition [7], and structural segmentation [8]—have decades of research history. These tasks are typically evaluated on dedicated datasets such as GTZAN [9], the Ballroom set [10], and the Beatles TUT [11], using task-specific metrics. Our benchmark repurposes these well-established tasks for the MLLM era: instead of evaluating a DSP algorithm, we evaluate whether a multimodal LLM can perform the same analysis that a DSP system does, but through natural language.

2.2 Audio-Language Benchmarks

MuChoMusic [1] is the closest prior work. It builds 1,187 multiple-choice questions over 644 tracks from two public datasets, evaluating five open-source audio-language models and finding an over-reliance on the language modality. Our benchmark differs in three key respects: (1) we focus on *structural* understanding (BPM, key, time signature, segmentation) rather than broad musical knowledge; (2) we *anonymize* all pieces to prevent title-based retrieval; and (3) we use *professional annotation* as the gold source rather than crowd-sourced or auto-generated labels.

2.3 Music Structure Analysis Datasets

Datasets such as SALAMI [12], SPAM [13], and the RWC Music Database [14] provide structural annotations. Our gold labels are derived from a commercial music annotation platform where professional annotators label segment boundaries and per-segment musical attributes (BPM, key, time signature, instruments, vocal type) as part of their standard workflow. This provides richer per-segment metadata than typical structure-only datasets.

2.4 Anonymization in Benchmarks

The concern that models may exploit metadata leakage rather than genuine understanding is well-documented in vision-language benchmarks [15]. In music, title-to-metadata leakage is particularly acute: a model trained on web data may have seen "Stairway to Heaven is

in A minor" and can answer key questions without listening. Our anonymization protocol (Section 3.4) directly addresses this threat model.

3 Dataset Construction

3.1 Source Corpus

The gold labels in MUB are derived from a professional music annotation platform where trained music analysts annotate commercial music as part of their standard workflow. Each annotation task involves:

- Listening to a complete music recording
- Segmenting the piece into structural parts (intro, verse, chorus, bridge, solo, outro, etc.)
- Labeling each segment with BPM (tempo), tonality (key and mode), time signature, and detailed instrument/vocal classifications
- Aligning lyrics to time intervals

The source corpus contains annotations for over 1,100 unique songs, each annotated by multiple analysts through multiple revision passes. Each annotation submission is timestamped and attributed, enabling provenance tracking.

3.2 Selection Criteria

From the source corpus, we apply the following filters to select benchmark candidates:

1. **Quality**: Only annotations that passed quality review are considered.
2. **Structural richness**: Pieces must have at least 5 annotated segments (≥ 5), ensuring sufficient structural complexity for meaningful segmentation evaluation.
3. **Duration**: 90–600 seconds, excluding very short clips (structurally unstable) and very long pieces (atypically complex).
4. **Unambiguous BPM**: The modal BPM across all segments must be a clear majority (no ties), and within 40–220 BPM.
5. **Unambiguous key**: The modal tonality across all segments must be a clear majority and parseable into a standard (pitch class, mode) pair.
6. **Unique songs**: Deduplicate by title and artist, keeping the version with the most segments. (No identifying information is retained in the released benchmark; this deduplication occurs before anonymization.)

After filtering, 1,111 unique songs qualify as candidates.

3.3 Stratified Selection

From the 1,111 candidates, we select 500 pieces using stratified sampling to ensure diversity:

- **BPM bins**: slow (47–69), medium (70–99), fast (100–129), very fast (130–180)
- **Tonality**: 24 key/mode combinations (12 pitch classes $\times$ 2 modes)
- **Time signature**: 4/4 (dominant), 3/4, 6/8

Strata are defined by (BPM bin, key/mode). We fill rare strata first (at least 1 piece per stratum), then proportionally fill from remaining candidates. This prevents over-sampling the modal C-major/120-BPM cluster and ensures coverage of unusual keys (e.g., F♯ minor, B major) and extreme tempos.

3.4 Anonymization Protocol

A central design principle of MUB is that **no identifying information** reaches the model. The anonymization protocol strips all fields that could enable title-based retrieval:

Field	In source	In MUB
Original title	Yes	**Removed**
Artist name	Yes	**Removed**
Platform song ID (metasong_id)	Yes	**Removed**
Internal task ID	Yes	**Removed**
Annotation account name	Yes	**Removed**
Operator names	Yes	**Removed**
Opaque benchmark ID	—	MUB-00001 … MUB-00500
Audio content hash (SHA-256, first 1 MB)	—	Included
Duration (seconds)	—	Included
Gold labels (BPM, key, time sig., segments)	—	Included
Annotation provenance (submission count, operator count)	—	Included (anonymized)

Table 1: Fields stripped and retained during anonymization.

The **audio content hash** (SHA-256 of the first 1 MB of the audio file) serves a dual purpose: (1) it allows participants to verify they have the correct audio file without us distributing the recording, and (2) it provides a non-identifying reference for audio provenance. We do not distribute audio files; participants must supply their own audio or use the provided DSP extraction pipeline.

3.5 Dataset Statistics

Table 2 summarizes the diversity of the 500 selected pieces. The BPM distribution spans four bins with roughly balanced coverage. All 24 key/mode combinations are represented, with 268 major and 232 minor keys. The time signature distribution is skewed toward 4/4 (98.4%), reflecting the popular-music bias of the source corpus.

Table 3 shows the distribution of structure labels in the benchmark. The canonical pop-music labels (chorus, verse, intro, outro, pre-chorus) dominate, but the vocabulary also includes electronic-music labels (build-up, breakdown, drop) and thematic labels (theme A/B/C/D).

4 Benchmark Tasks

MUB evaluates four tasks that collectively assess a model's ability to perceive and understand the structural and analytic dimensions of music. These tasks mirror the workflow of a professional music analyst and are designed to require genuine audio listening—they cannot be answered from text alone.

4.1 Task 1: Tempo Estimation (BPM)

Prompt. "Estimate the tempo in beats per minute (BPM) of this music."

Property	Value
Total pieces	500
BPM range	47–180
BPM: p25 / median / p75	75 / 101 / 128
Slow (47–69)	87
Medium (70–99)	156
Fast (100–129)	150
Very fast (130–180)	107
Unique keys (key + mode)	24
Major / Minor	268 / 232
Time signatures	3
4/4	492
3/4	4
6/8	4
Duration range	92–535 s
Duration median	228 s
Segments per piece	5–22
Segment median	9
Total annotated segments	4,676
Unique structure labels (lowercased)	34

Table 2: MUB dataset statistics. BPM is well-distributed across four bins; all 24 key/mode combinations are represented.

Label	Count	Label	Count	Label	Count
chorus	1231	interlude	324	theme c	21
verse	1114	bridge	282	breakdown	20
intro	460	post-chorus	154	hook	16
outro	427	solo	93	drop	15
pre-chorus	392	theme a	33	*+ 14 others*	
		theme b	26	build-up	28

Table 3: Structure label distribution (lowercased, top 16 of 34).

Gold. The modal BPM across all annotated segments, as labeled by expert annotators who listened to the recording and tapped or counted the beat.

Rationale. BPM is a physical property of the audio waveform, directly measurable by DSP methods. A model that cannot estimate BPM is not perceiving the basic rhythmic pulse of the music. This task tests the lowest level of audio understanding: temporal periodicity detection.

4.2 Task 2: Key and Tonality Identification

Prompt. "Identify the musical key (tonality) of this piece. Respond with the tonic pitch class and the mode (major or minor)."

Gold. The modal tonality across all segments, normalized to a (pitch class, mode) pair. Both the absolute key (e.g., "C major") and the mode (major vs. minor) must be correct.

Rationale. Key identification requires perceiving the tonal center and the major/minor modality—a fundamental dimension of Western music. This cannot be determined from lyrics or genre; it

requires hearing the pitch content and harmonic relationships.

4.3 Task 3: Time Signature Detection

Prompt. "Identify the time signature of this piece (e.g., 4/4, 3/4, 6/8)."

Gold. The modal time signature across all segments.

Rationale. Time signature reflects the metrical grouping of beats (duple vs. triple, simple vs. compound). While 4/4 dominates popular music, distinguishing 3/4 and 6/8 requires perceiving the pattern of strong and weak beats, not just the tempo.

4.4 Task 4: Structural Segmentation

Prompt. "Segment this music into its structural parts (e.g., intro, verse, chorus, bridge, outro, solo). For each segment, provide the start time, end time, and a label."

Gold. Expert-annotated segment boundaries with descriptive labels (e.g., "Intro", "Verse 1", "Chorus", "Guitar Solo", "Outro").

Rationale. Structural segmentation is the most cognitively demanding task: it requires the model to identify where sections begin and end based on changes in texture, instrumentation, harmony, and dynamics. This is a *semantic* task that DSP methods cannot perform (they can detect novelty boundaries but cannot label segments). It directly tests whether the model can build a hierarchical understanding of the music's form.

5 Evaluation Metrics

5.1 BPM Estimation

BPM is scored as a binary hit if the predicted value is within ±4% of the gold value:

$$\text{BPM hit} \iff |\text{BPM}_{\text{pred}} - \text{BPM}_{\text{gold}}| \leq 0.04 \times \text{BPM}_{\text{gold}} \tag{1}$$

The 4% tolerance reflects music-engineering practice: tempos within 4% are perceptually indistinguishable and within the variation of human tapping. We also report the median relative error for continuous assessment.

5.2 Tonality Identification

Tonality is scored as a hit if the predicted (pitch class, mode) matches the gold, *or* if they are relative major/minor (same pitch class set, 3-semitone offset, different mode). Relative major/minor keys share the same key signature and are perceptually close; treating them as equivalent follows standard MIR practice.

5.3 Time Signature Detection

Scored as an exact match after normalization ("4/4" matches "4 / 4" and "4,4"). Given the small number of possible values, partial credit is not applied.

5.4 Structural Segmentation

Segmentation is scored using a greedy IoU-based matching protocol:

1. For each gold segment, find the best-IoU unmatched prediction.
2. If IoU ≥ 0.5, count it as a match.
3. Compute precision $P = \text{matched}/|\text{pred}|$, recall $R = \text{matched}/|\text{gold}|$, and $F_1 = 2PR/(P+R)$.

Greedy matching (rather than Hungarian assignment) is sufficient given the small number of segments (5–22 per piece). Note that this metric evaluates boundary detection; label accuracy is assessed separately as a secondary metric.

6 Baselines

6.1 Objective DSP Baseline

The DSP baseline extracts musical features directly from the audio waveform, without any language model or semantic understanding. It represents "a system that can truly hear the audio but understands no meaning."

BPM. Spectral flux (half-wave-rectified frame-to-frame spectral difference) is computed via STFT with a 2048-point window and 512-sample hop at 22050 Hz. The autocorrelation of the flux signal reveals the periodicity corresponding to the beat. The BPM is estimated from the autocorrelation peak in the 60–180 BPM range, folding half/double tempo errors.

Key. A chroma vector (12-dimensional pitch-class energy) is computed by mapping FFT bins to pitch classes via MIDI note numbers. The Krumhansl–Schmuckler key profiles [16] are correlated against the chroma vector for all 24 keys (12 tonics $\times$ 2 modes); the best-correlated key is selected.

Time signature. The beat interval (from BPM) is used to test periodicity at 2, 3, and 4 beat multiples. The strongest autocorrelation peak among these determines the numerator; the denominator is set to 4 (common practice for the tested meters).

This baseline serves two purposes: (1) as a *lower bound* on achievable accuracy with genuine audio access, and (2) as a *credibility cross-check*—if the DSP baseline agrees with expert annotations on BPM, it validates that the gold labels are genuinely derived from listening.

6.2 Majority-Vote Baseline

The majority baseline predicts the single most common value across the benchmark for every piece: the median BPM (101), the modal key (C major), the modal time signature (4/4), and no structure prediction. It simulates a model that has no audio access and no music knowledge—only knowledge of the distribution of answers in the dataset. Any model that fails to beat this baseline has learned nothing from the audio input.

6.3 Baseline Results

Table 4 shows the majority baseline results on the full 500-piece benchmark. The results confirm expectations: without audio, the model can only guess the modal answer, achieving low accuracy on BPM and key. Time signature accuracy is deceptively high (98.4%) because 4/4 dominates the dataset. Structure F1 is zero because the majority baseline cannot produce segment predictions.

Task	Metric	Majority Baseline	Random Chance
BPM estimation	Accuracy (±4%)	9.0%	∼1.3%
Tonality	Accuracy	9.2%	∼4.2%
Time signature	Accuracy	98.4%	∼33.3%
Structure	Mean F1	0.000	0.000

Table 4: Majority-vote baseline results on the full 500-piece benchmark. "Random chance" assumes uniform guessing over the observed value space.

7 Analysis and Discussion

7.1 Expected Performance Gaps

Based on preliminary DSP baseline results from the source corpus, we anticipate the following performance landscape for MLLMs:

- **BPM**: The DSP baseline achieves ∼56% accuracy (within ±4%) with a median relative error of ∼2.5% on the source corpus. This confirms that expert annotations are genuinely audio-derived. A multimodal LLM that cannot match DSP performance is not perceiving rhythm.
- **Key**: The DSP baseline achieves ∼28% accuracy—low, but significantly above the majority baseline (∼1/24 ≈ 4%). Key estimation from chroma is inherently noisy for complex polyphonic music, making this a challenging task even for dedicated algorithms.
- **Time signature**: The majority baseline already achieves 98.4% by always predicting "4/4" (since 98.4% of the dataset is 4/4). This makes time signature detection a *low-information task* in our current dataset; future versions should oversample non-4/4 meters.
- **Structure**: The DSP baseline achieves $F_1 = 0$ (it cannot label segments). The majority baseline also cannot produce structure. This task is the *purest test of semantic audio understanding*—only a model that truly comprehends musical form can score above zero.

7.2 The Title-Retrieval Threat

A fundamental motivation for MUB's anonymization is the title-retrieval shortcut. Consider a model that has seen "Stairway to Heaven is in A minor" in its training data. Without anonymization, it can answer the key question correctly without listening to a single note—but this proves nothing about its audio understanding ability. By stripping all identifying information, MUB ensures that the only path to a correct answer runs through the audio signal.

7.3 The Tool-Use Threat and Prompt Constraints

Anonymization alone is insufficient if the evaluated model has access to external tools. A model with internet access could, in principle, use audio fingerprinting services (e.g., Shazam-like APIs), music recognition databases, or web search to identify the recording from its audio content—bypassing the anonymization entirely and converting a perception task into a database lookup.

To close this loophole, every MUB prompt includes an explicit constraint preamble prohibiting such behavior:

> *CONSTRAINTS — Read before answering:*
>
> *1. Base your answer solely on what you hear in the provided audio.*

2. *Do NOT use web search, internet access, or any external tool to identify the recording.*
3. *Do NOT attempt to use audio fingerprinting, music recognition services, or any external database lookup.*
4. *Do NOT use any prior knowledge about what this recording might be—the piece has been deliberately anonymized.*
5. *Respond based only on your own audio perception and musical analysis.*

We note that prompt-based constraints are not a guarantee: a model that ignores instructions or a deployment that auto-routes to external tools could still violate them. However, for the intended use case—evaluating a model's native multimodal perception in a controlled API setting—these constraints, combined with anonymization, close the two most obvious shortcuts: title-based retrieval and tool-based identification. Researchers using MUB should ensure their evaluation harness disables any tool-calling or web-search capabilities before running the benchmark.

7.4 Limitations

- **Genre bias**: The source corpus is weighted toward popular music (pop, rock, electronic), reflecting the annotation platform's commercial focus. Classical, jazz, and world music are underrepresented.
- **Time signature imbalance**: 98.4% of pieces are in 4/4, limiting the discriminative power of the time signature task. Future work should actively recruit non-4/4 pieces.
- **Label vocabulary**: Structure labels exhibit capitalization inconsistencies ("chorus" vs. "Chorus") and free-text variation, which the evaluation pipeline normalizes but which may affect label-level scoring.
- **No audio distribution**: We do not distribute audio files due to copyright constraints. Participants must supply their own audio, which creates a reproducibility barrier. The audio hash enables verification but not direct access.

8 Conclusion

We have presented MUB, a fully anonymized benchmark for evaluating multimodal LLMs on music structural understanding. By stripping all identifying information and grounding gold labels in professional annotation, MUB provides a clean test of whether a model can truly *listen* to music. The four tasks—BPM estimation, key identification, time signature detection, and structural segmentation—span from low-level temporal periodicity to high-level semantic form analysis, enabling granular diagnosis of model capabilities.

The DSP and majority baselines establish critical reference points: the DSP baseline proves that the gold labels are audio-derived and sets a perceptual lower bound; the majority baseline exposes tasks where distributional guessing suffices. Together, they allow researchers to distinguish genuine audio understanding from prior knowledge artifacts.

We release the benchmark, evaluation code, and baselines as an open repository, inviting the community to evaluate current and future MLLMs on music understanding.

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